



FACULTY OF COMPUTING
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SECJ 3553 - ARTIFICIAL INTELLIGENCE
SECTION 03

AI Project - Assignment 2

THEME:
SMART CITY - ECO-ZONE AIR CONDITIONER

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1.0 Problem Formulation by State Space Graph

The state space graph below shows the possible temperature, conditions and actions that the Eco-Zone Air Conditioner system may encounter. It guides the searching for the most optimal path based on the library internal and external conditions while improving users' comfort and energy sustainability.

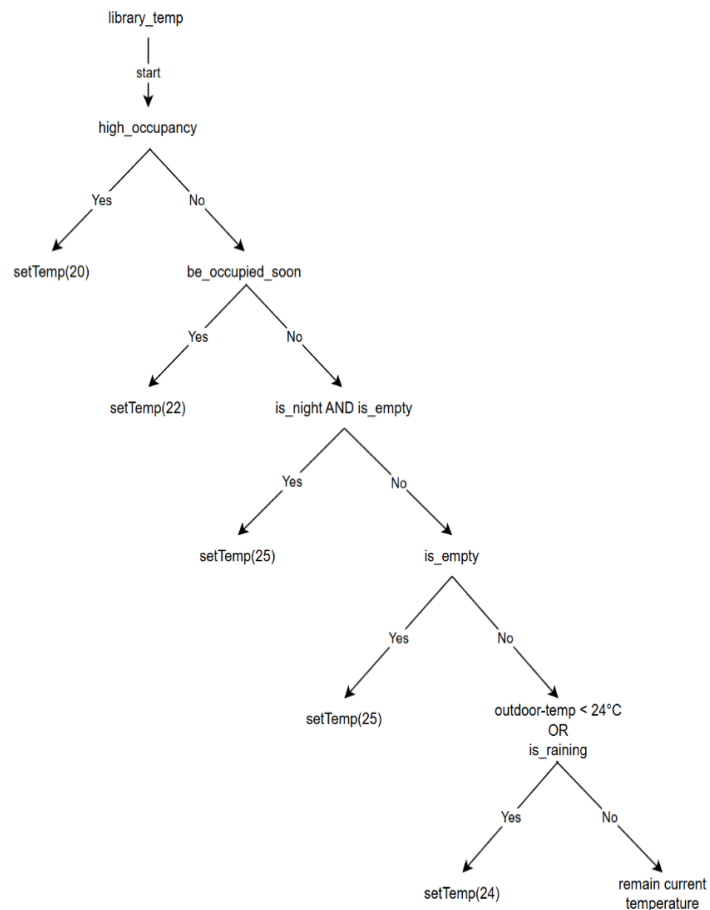


Figure 1.0 Eco-Zone Air Conditioner State Space Graph

1.1 States Explanation

State	Explanation
is_empty	the state where the library is checked to have no occupants.
high_occupancy	the state where the library is checked to have many occupants.
be_occupied_soon	the state where the system predicts the library will have occupants soon.
is_night	the state where the current time is checked to be during nighttime AND is after 10p.m.
is_raining	the state where the weather is checked for rain.
outdoor-temp < 24°C	the state where the outdoor temperature is checked to be below 24°C.
library_temp	the state where the current temperature inside the library is checked.
remain current temperature	the state where the system keeps the library's temperature unchanged.

2.0 Problem Formulation and Explanation

The air conditioning automated control scenario is formulated as a search problem within a State Space Graph where the system navigates through sensor data to arrive at an optimal action or temperature setting.

2.1 Initial State

The initial state in the graph is the start node which represents the status of the library_temp (the current air conditioner setting within a specific zone). The system transitions from this initial state by evaluating a series of condition-based nodes, which are the fundamental states of the environment:

State Node	Condition Checked	Explanation
high_occupancy	Occupancy is 10 or more people.	The first and highest priority check, intended to ensure immediate user comfort by addressing high heat load.
be_occupied_soon	The predictive model forecasts zone occupancy at future time.	This check facilitates proactive pre-cooling for comfort before users arrive.
is_night AND is_empty	Current time is night-time (after 10p.m.) AND the zone is empty.	A specific joint state targeting maximum energy savings during low-usage, late hours.
is_empty	The motion sensor detects no occupants.	A general check for standard energy saving when the zone is unused

		during normal operating hours.
outdoor-temp < 24°C OR is_raining	External temperature is low OR rain is detected.	This final condition checks if outdoor weather can assist cooling, allowing for an energy-efficient temperature adjustment while the zone is occupied.

2.2 Actions

The actions are the terminal nodes of the search, representing the specific temperature command issued to the AC unit (setTemp(T)) or a command to maintain the current setting.

Action	Objective
setTemp(20)	Triggered by high_occupancy to aggressively counteract body heat.
setTemp(22)	Used for predictive cooling (be_occupied_soon).
setTemp(24)	Used when outdoor temp < 24°C OR is_raining to conserve energy using external factors.
setTemp(25)	Used when the zone is is_empty (both night and day checks) to minimize power consumption.
remain current temperature	The output when the zone is occupied, and no specific high-priority comfort or efficiency conditions are met, preventing unnecessary cycling.

2.3 Goal

Optimal Temperature Control

The goal is to obtain the ideal temperature setting that balances user comfort and energy efficiency using real-time sensor data (occupancy, weather, time) and predictive analytics. Based on an analysis of the various conditions detected by real-time sensors, the system selects one of five different temperature control strategies which is 20°C for maximum comfort in high-occupancy situations, 22°C for prediction-based pre-cooling, 24°C for weather adaptive cooling, 25°C for energy conservation when all zones are unoccupied and maintaining the current room temperature during periods of no external factors that may require adjustment.

2.4 Path Cost

Path	Solution Route	Path Cost
1	library_temp, high_occupancy, setTemp(20)	2 units
2	library_temp, high_occupancy, be_occupied_soon, setTemp(22)	3 units
3	library_temp, high_occupancy, be_occupied_soon, is_night AND is_empty, setTemp(25)	4 units
4	library_temp, high_occupancy, be_occupied_soon, is_night AND is_empty, is_empty, setTemp(25)	5 units
5	library_temp, high_occupancy, be_occupied_soon, is_night AND is_empty, is_empty, outdoor-temp < 24°C OR is_raining, setTemp(24)	6 units
6	library_temp, high_occupancy, be_occupied_soon, is_night AND is_empty, is_empty, outdoor-temp < 24°C OR is_raining, remain current temperature	6 units

In this case, the path cost means the total time consumed by the system to reach the goal of perfect temperature control through the orderly assessment of sensor data and environmental factors. The cost is qualified by the amount state transition which is the condition checks done in each solution path, with each state evaluation indicating one unit of processing time. Paths with less evaluation have smaller cost and receive higher priority for critical comfort scenarios while paths with more evaluation show thorough condition checking for energy for energy optimization.